Polynomial-depth Bayesian learning with general Lipschitz activations under balanced neuron-averaging stability

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Abstract

We give one polynomial sample-complexity argument for tanh, ReLU, exact GELU, SiLU, sigmoid, softplus, and every fixed globally Lipschitz activation. No zero-at-zero assumption is needed. The Gaussian prior is restricted to networks whose clipped predictions survive averaging neurons within sufficiently few balanced populations, with population count calibrated by readout energy. A target implemented by a depth-L, width-C network with weights bounded by B is learned with sample complexity polynomial jointly in d, C, L, B, the activation's Lipschitz and offset bounds, and the statistical parameters. One sample event works at every learner width $N \geq C$ and every prior scale at least as strong as mean field, including all $\gamma \geq 1$. The teacher need not itself satisfy the restriction: a rescaled, cloned implementation has a neighborhood of sufficient restricted prior mass. Its hidden states may grow exponentially with depth; the resulting small parameter radius costs only its logarithm. This is a theorem for a structurally restricted prior, not for the unrestricted Gaussian learner or for internal-noise stability alone. A separate appendix removes the zero-at-zero assumption from the fixed-depth learning theorem for the original globally margin-conditioned prior. The proofs are conventional and are not Lean formalized.

1 The result and its setting

The question is whether a single argument can give polynomial dependence on depth for Bayesian networks with both bounded and unbounded activations. We use a restriction expressing redundancy among hidden neurons. The same balanced-population restriction was previously studied for tanh networks; we show that its proof applies much more broadly. In particular, neither positive homogeneity nor $\phi(0) = 0$ is needed. The comparator here is an actual bounded-weight neural teacher, so no Boolean-circuit compiler or extra constant wire is hidden in the width threshold.

Fix $d, C, H \geq 1$, $B, \lambda \geq 1$, and $b \geq 0$. Network depth is $L = H + 1$. Fix any activation $\phi : \mathbb{R} \to \mathbb{R}$ such that

$$|\phi(u) - \phi(v)| \leq \lambda |u - v|, \qquad |\phi(0)| \leq b. \tag{1}$$

Inputs are $x \in \mathcal{X} = \{-1, 1\}^d$, with $\bar{x} = (x, 1)$. A width-N network has H hidden layers and a linear readout:

$$h^1 = \phi(W^1 \bar{x}), \qquad h^s = \phi(W^s h^{s-1}) \quad (2 \leq s \leq H), \qquad z_w = v^\top h^H.$$

Activation is coordinatewise; there are no separate internal biases. The raw prior $\mu_{N,t}$ has independent centered Gaussian entries with variances

$$t^2/d \text{ in } W^1, \qquad t^2/N \text{ in } W^s \ (s \geq 2), \qquad t^2/N^2 \text{ in } v, \qquad 0 < t \leq 1. \tag{2}$$

The original exponent convention is $t = N^{(1-\gamma)/2}$, so this interval covers every $\gamma \geq 1$. Define

$$E_v = N \|v\|_2^2, \qquad T(u) = \max\{-1/2, \min\{1/2, u\}\}.$$

A balanced partition into q populations has sizes in $\{\lfloor N/q \rfloor, \lceil N/q \rceil\}$. Let P_s average the coordinates within the populations at hidden layer s. For a tuple of such partitions, all with the same q, define

$$\bar{W}^1 = P_1 W^1, \qquad \bar{W}^s = P_s W^s P_{s-1}, \qquad \bar{v} = P_H v. \tag{3}$$

Let $\mathcal{A}_N^\phi$ be the event that some q and balanced partitions satisfy

$$1 \leq q \leq N, \qquad q \leq 1 + E_v(w), \qquad \max_{x \in \mathcal{X}} |T(z_w(x)) - T(z_{\bar{w}}(x))| \leq 1/16. \tag{4}$$

This event depends on the activation and network, but not on the target or training labels. It is Borel: there are finitely many witnesses at fixed N, and the output maps are continuous. The condition uses absolute readout energy; it is not invariant under arbitrary rescaling of the readout.

The prior is $\mu_{N,t}$ conditioned on $\mathcal{A}_N^\phi$. We also allow the optional further restriction $|z_w(x)| > 1/2$ at every input. Training conditions on $f(X_i) z_w(X_i) > 1/2$ for every example, where X_i are iid from any distribution D on the cube. Write $Q_{S,N,t}$ for the resulting posterior, with either convention. Test loss is

$$R_D(Q) = \mathbb{E}_{w \sim Q} \Pr_{X \sim D} \{f(X) z_w(X) \leq 1/4\}.$$

This is a full posterior-average margin loss, which bounds sign error.

Theorem 1 (One polynomial for all activations and all stronger scales)**.** *There is a universal constant K_0 with the following property. Suppose $f : \mathcal{X} \to \{-1, 1\}$ has a width-C, H-hidden-layer implementation using ϕ, all of whose individual weights are bounded by B, and whose output satisfies $f(x) z_*(x) > 1/2$ at every input. Put*

$$S_0 = 2 + d + C + H + B + \lambda + b + \log(4/\varepsilon).$$

For every $0 < \varepsilon, \delta < 1$, the sufficient condition

$$n \geq \frac{K_0}{\varepsilon} \left[S_0^{13} + \log(1/\delta) \right] \tag{5}$$

gives, with sample probability at least $1 - \delta$, $R_D(Q_{S,N,t}) \leq \varepsilon$ simultaneously for every $N \geq C$, every $0 < t \leq 1$, and both choices of the optional global-margin restriction. All prior and training normalizers are positive. The sample polynomial does not depend on N, t, or γ, and its coefficient and degree are uniform over activations satisfying (1).

The sample event may depend on the fixed activation and target. The statement does not select an activation using the same data. The teacher uses the learner's activation; no arbitrary Boolean circuit is identified with a same-width implementation. Under the earlier complexity convention $B = C$, the displayed envelope is still jointly polynomial in width and depth. Its degree is deliberately conservative, not claimed optimal.

Corollary 2 (Posterior confidence)**.** *Apply Theorem 1 with accuracy $\varepsilon/3$. Then the posterior probability of $f(x) z_w(x) > 1/4$ is strictly greater than $2/3$ outside a set of test mass at most ε. This also implies the same guarantee for the correct sign. The conclusion allows arbitrary posterior mixtures.*

Proof. If the correct-margin probability is at most $2/3$, the posterior error at that input is at least $1/3$. Integrate this inequality under D. □

2 Explicit budgets

The following formulas make the polynomial statement concrete. Define

$$
\begin{aligned}
R_* &= 2 + b + \lambda B(d + C + 1), & M &= R_*^H, \\
g_H &= 2 + \sqrt{2\log(2H)}, & A_* &= 1 + \lambda(\sqrt{2}CB + \sqrt{C} + g_H), \\
V &= 4\sqrt{2}CB, & E &= \lambda(d + 1 + H\sqrt{C})MA_*^{H-1}, \\
\rho &= \frac{1}{8[(V+1)E + M]}, & B_0 &= B^2[d(d+1) + (2H+30)C^2], \\
q_0 &= d + 2 + (H-1)C, & D_* &= B_0 + q_0 \log\frac{e\sqrt{2\pi}}{2\rho} + \log 2.
\end{aligned}
$$

The potentially exponential quantities $M, E, 1/\rho$ enter the prior cost only through $\log(1/\rho)$. Set

$$
B_r = \max\{8, 8[D_* + \log(4/\varepsilon)]\}, \qquad Q = \lceil 1 + B_r \rceil.
$$

$$
J = 8\left[D_* + \log(4/\varepsilon) + (H+1)\log Q + \tfrac{\log 2}{2}\{d + 2 + (H-1)Q\}\right].
$$

$$
\begin{aligned}
R &= 1 + \sqrt{QJ}, & A_g &= 2(1 + b + \lambda)(d + 1 + Q)(R+1), \\
\zeta &= [32(H+1)A_g^{H+1}]^{-1}, & P_Q &= Q(d+2) + (H-1)Q^2.
\end{aligned}
$$

Theorem 3 (Explicit sufficient sample size)**.** *Every conclusion of Theorem 1 holds if*

$$
n \geq \frac{2}{\varepsilon}\left[P_Q \log(3 + 4R/\zeta) + \log(1/\delta)\right]. \tag{6}
$$

3 A target neighborhood of polynomial logarithmic cost

Lemma 4 (Target mass)**.** *At every $N \geq C$ and $0 < t \leq 1$, the raw Gaussian mass of networks in $\mathcal{A}_N^\phi$ satisfying $f(x)z_w(x) > 1$ everywhere is at least* $\exp(-ND_*/t^2)$.

Proof. Let v be the teacher readout. Its norm is positive by the signed-margin assumption. Replace it by

$$
v' = 4\max\{1, \|v\|_2^{-1}\}v.
$$

Then $4 \leq \|v'\|_2 \leq 4B\sqrt{C}$, and the signed output is greater than 2. The multiplier itself can be large; the final coefficient norm, which determines the Gaussian cost, is polynomially bounded.

Partition all N neurons at each layer into C balanced groups with sizes k_a. Copy first-layer rows, put an internal teacher weight from logical neuron a to b on each corresponding edge divided by k_a, and divide each readout coefficient by its group's size. This preserves every hidden value and the output exactly for every activation. The cloned hidden state is constant within groups, and its coordinates are bounded by M: the recurrence $m_1 \leq b + \lambda B(d+1)$, $m_s \leq b + \lambda CBm_{s-1}$ gives $m_s \leq R_*^s$.

For these cloned weights w_*,

$$
\|W_*^s\|_{\mathrm{op}} \leq \sqrt{2}CB \quad (s \geq 2), \qquad 8C \leq N\|v_*\|_2^2 \leq 32C^2B^2, \qquad \sqrt{N}\|v_*\|_2 \leq V.
$$

These follow from $C/2 \leq N/k_a \leq 2C$ and $k_b/k_a \leq 2$. In independent Gaussian coordinates $\sqrt{d}W^1$, the coordinates of each $\sqrt{N}W^s$ row along the C normalized group indicators, and Nv, constrain

every coordinate within ρt of its center. There are Nq_0 such coordinates, each with variance t^2, and the sum of squared centers is at most NB_0.

Writing P for the clone projection, an internal perturbed matrix is $W^s = W^s_* + F^s + G^s$ with

$$F^s = F^s P, \quad \|F^s\|_{\mathrm{op}} \le \rho t\sqrt{C}, \qquad G^s P = 0.$$

The free bulk $G^s = (t/\sqrt{N})Z^s(I-P)$ is independent of the constrained coordinates. The Gaussian operator-norm bound gives $\Pr\{\|G^s\|_{\mathrm{op}} > g_H t\} \le 1/(2H)$. Thus all $H-1$ bulk bounds hold with conditional probability at least $1/2$; for $H=1$ the assertion is immediate.

Put $e_s = \sup_x \|h^s_w(x) - h^s_*(x)\|_2 / \sqrt{N}$. Lipschitz continuity, $G^s h^{s-1}_* = 0$, and $t, \rho \le 1$ give

$$e_1 \le \lambda\rho(d+1), \qquad e_s \le A_* e_{s-1} + \lambda\rho\sqrt{C}M.$$

Consequently $e_H \le \rho E$. The scaled readout error is at most ρ, so the output error is at most

$$(V+1)e_H + \rho M \le 1/8.$$

The same estimate holds for the merged weights: the projections preserve the constrained coordinate intervals, the centers, and kill the free bulk. Both outputs therefore have signed value greater than $15/8$ everywhere. Their clipped outputs agree exactly. Also

$$\sqrt{E_v(w)} \ge \sqrt{8C} - \rho \ge \sqrt{2C},$$

since $\rho \le 1/8$ and $C \ge 1$. The clone partition thus witnesses $C \le 1 + E_v(w)$ and (4). The optional global margin holds too.

For a scalar $Z \sim \mathcal{N}(0, t^2)$ and $\rho \le 1$,

$$\Pr\{|Z-a| \le \rho t\} \ge \frac{2\rho}{e\sqrt{2\pi}} e^{-a^2/t^2}.$$

Multiply these interval bounds and the independent bulk probability. Using $t \le 1$ and $N/t^2 \ge 1$ yields the claimed lower bound. □

4 Gaussian control of every balanced partition

For fixed partitions, the merged network is exactly a width-q network with the original activation and coefficients

$$U^1_{aj} = \frac{1}{k_{1,a}} \sum_{i \in a} W^1_{ij}, \qquad U^s_{ab} = \frac{1}{k_{s,a}} \sum_{i \in a, j \in b} W^s_{ij}, \qquad u_a = \sum_{j \in a} v_j.$$

This identity holds for every activation, including nonzero offsets. Its weighted energy is

$$\kappa_q = \frac{d}{q} \left\| U^1 \right\|^2_F + \sum_{s=2}^{H} \|U^s\|^2_F + q \|u\|^2_2 .$$

Lemma 5 (Partition tail)**.** *The raw probability that some balanced partitions with $q \le \min\{N, Q\}$ have $\kappa_q > J$ is at most*

$$Q \exp\left[HN \log Q + \tfrac{N}{2}\{d + 2 + (H-1)Q\} \log 2 - \frac{NJ}{8t^2} \right].$$

Proof. For a fixed partition, compressed coordinates are independent centered Gaussians of variances $t^2/(dk_{1,a})$, $t^2k_{s-1,b}/(Nk_{s,a})$, and $t^2k_{H,a}/N^2$. After the energy factors, each variance is at most $2t^2/N$ by balance. There are $P_q = q(d+2) + (H-1)q^2$ coordinates, hence

$$\mathbb{E}e^{N\kappa_q/(8t^2)} \le 2^{P_q/2}.$$

There are at most q^{HN} ordered partition tuples, and $P_q/N \le d+2+(H-1)q$ for $q \le N$. Markov's inequality and the union bound prove the assertion. The union is taken before choosing any partition depending on the weights. □

Lemma 6 (Uniform posterior truncation)**.** *For every noiseless training sample, outside posterior mass at most $\varepsilon/2$, a witnessing merged network has $q \le Q$ and $\kappa_q \le J$, for every eligible width, scale and optional-margin choice.*

Proof. The raw mass of each conditioning event is at least e^{-ND_*/t^2} by Lemma 4. Since NE_v/t^2 is chi-square with N degrees of freedom, for $B_r \ge 8$,

$$\mu_{N,t}\{E_v > B_r\} \le e^{-NB_r/(4t^2)}2^{N/2} \le e^{-NB_r/(8t^2)}.$$

After normalization this is at most $\varepsilon/4$. Otherwise support membership supplies $q \le 1 + B_r \le Q$. Apply Lemma 5. Each positive exponent term, including $\log Q$, is bounded by N/t^2 times its corresponding term in $J/8$. The remaining raw probability is at most $e^{-ND_*/t^2}(\varepsilon/4)^{N/t^2}$; after normalization it is at most $\varepsilon/4$. Both estimates use only $N \ge 1$ and $t \le 1$. □

5 Finite coding for activations with an offset

A merged network with $q \le Q$ and energy at most J has coefficients bounded by $\sqrt{QJ}$. Pad to width Q, assigning zero to all edges entering or leaving a dummy neuron and to its readout coefficient. Dummy hidden outputs equal $\phi(0)$, but their outgoing weights are zero, so padding preserves the output exactly. Round coefficients on a fixed grid with mesh error at most ζ, preserving zero, and magnitude bounded by R. The class $\mathcal{G}$ of all grid networks has

$$\log|\mathcal{G}| \le P_Q \log(3 + 4R/\zeta). \tag{7}$$

Every network in this coding step still uses the one fixed activation.

The inequality $|\phi(u)| \le b + \lambda|u|$ gives hidden-coordinate magnitudes at most A_g^s. If e_s is the largest coordinate difference between the padded and rounded networks, then

$$e_1 \le \lambda(d+1)\zeta \le A_g\zeta, \qquad e_s \le \lambda QRe_{s-1} + \lambda Q\zeta A_g^{s-1} \le A_g e_{s-1} + \zeta A_g^s.$$

Thus $e_s \le s\zeta A_g^s$, and the readout error is at most $(H+1)\zeta A_g^{H+1} = 1/32$. Merging and clipping give

$$|T(z_w(x)) - T(g(x))| \le 3/32 \tag{8}$$

for some $g \in \mathcal{G}$, uniformly over inputs. Only the logarithm of the exponential precision enters (7).

Proof of Theorem 3. For each grid predictor define $F_g = \{x : f(x)T(g(x)) \le 11/32\}$. If the original network fits the training margins and satisfies (8), then $f(X_i)T(g(X_i)) \ge 13/32$ for every training input, so the sample avoids this event. Every test failure $f(x)z_w(x) \le 1/4$ belongs to F_g. A fixed event of probability greater than $\varepsilon/2$ is avoided by n iid inputs with probability at most $e^{-n\varepsilon/2}$. Union bounding over (7) gives one sample event of probability at least $1-\delta$ on which every retained network has test loss at most $\varepsilon/2$. The discarded posterior mass from Lemma 6 adds at most $\varepsilon/2$. The grid is independent of N, t and the optional-margin choice, proving simultaneity. Lemma 4 proves all normalizers positive, even without a lower bound on sample size. □

Proof of Theorem 1. All constants in the following estimates are universal. For S_0 as in the theorem, the target budgets satisfy

$$R_* \le 2S_0^3, \quad A_* \le 8S_0^3, \quad V \le 6S_0^2, \quad \log(1/\rho) = O(H \log S_0), \quad D_* = O(S_0^5).$$

The last bound follows from $B_0 = O(S_0^5)$, $q_0 = O(S_0^2)$, and $H \log S_0 = O(S_0^2)$. Consequently

$$Q = O(S_0^5), \quad J = O(S_0^6), \quad R = O(S_0^6), \quad A_g = O(S_0^{12}), \quad P_Q = O(S_0^{11}).$$

Finally $\log(3 + 4R/\zeta) = O(H \log S_0) = O(S_0^2)$, so the log-code budget is $O(S_0^{13})$. Increase one universal constant to dominate the explicit sample condition. No exponent depends on depth or activation. □

6 Scope and interpretation

The activation class includes all the examples below, with the indicated conservative resource bounds.

Activation	Lipschitz bound λ	Offset bound b
$\tanh(u)$, ReLU	1	0
Exact GELU $u\Phi(u)$, SiLU $u/(1+e^{-u})$	2	0
Sigmoid $(1+e^{-u})^{-1}$	1	1/2
Softplus $\log(1+e^u)$	1	$\log 2$

For a fixed activation its offset is a fixed constant. Removing $\phi(0) = 0$ therefore admits ordinary sigmoid and softplus without a width penalty. Simply centering an activation is not an equality of the original Bayesian learners: it creates hidden biases correlated with the weights. Our argument analyzes the uncentered learner directly.

The prior restriction remains substantial. It requires balanced hidden populations and prediction stability under their averaging, with the count-energy relation (4). It is not implied here by small internal-noise sensitivity, a bounded Jacobian, ordinary training, or the mere existence of a small teacher. Its positive-mass teacher construction establishes representability under the restriction, not its prevalence among trained networks. The theorem does not retain an additional arbitrary internal-noise restriction for general analog teachers. Such an intersection needs its own target-mass argument.

This condition differs from the unsigned, arbitrary-population-size, output-scale-invariant restriction in the ReLU/GELU polynomial-depth paper. Balance bounds ratios of population sizes and lets us code the original fixed activation directly; that is why no special perspective identity for GELU, or homogeneity for ReLU, is needed. The sharper quadratic Boolean-width results under those other conditions remain separate. Nor does the present result settle unrestricted polynomial-depth learning, efficient posterior inference, corrupted labels, or arbitrary real-valued regression.

Relation to earlier work and verification. The balanced-population Gaussian tail and finite-code argument are inherited from the collaborative tanh population paper, with contributions credited there to ChatGPT-6 Astra and Claude. The new points here are the general bounded-weight analog-teacher cylinder, its readout normalization, and uniform coding for offset Lipschitz activations. These remove both the digital-teacher restriction and the activation-specific parts of the proof. The present extension has conventional proofs and AI-agent checks; the earlier Lean certifications do not automatically certify it.

A Nonzero activation offsets for the original prior at fixed depth

The restriction (4) is unnecessary if depth is held fixed and its polynomial exponent is allowed to depend on depth. This appendix extends the unified fixed-depth theorem of [3], whose stated activation assumption was $\phi(0) = 0$, to arbitrary bounded activation offsets. It analyzes the original Gaussian learner directly. Centering an activation and declaring the resulting network to have the same prior would be incorrect, since it induces weight-dependent hidden biases. The result below is a conventional proof extension; the earlier Lean certification does not certify this new offset extension.

Keep the network and Gaussian law (2), but impose *no internal-noise or neuron-averaging support restriction.* Instead, let

$$U_{N,g} = \{w : g(x)z_w(x) > 1/2 \text{ for every } x \in \mathcal{X}\}, \qquad U_N = \bigcup_g U_{N,g}.$$

The prior is $\mu_{N,t}$ conditioned only on U_N. Given noiseless examples of f, retain functions consistent with all training labels; write the resulting function posterior as $\Pi_{S,N,t}$. Its risk is

$$R_D^{\mathrm{sign}}(\Pi) = \mathbb{E}_{g\sim\Pi} \Pr_{X\sim D}\{g(X) \neq f(X)\}.$$

Unlike the main theorem's margin loss, this is sign error under a prior already conditioned on the global absolute margin.

Theorem 7 (Original-prior learning with bounded activation offsets)**.** *For each fixed $L \geq 2$, $\lambda \geq 1$ and $b \geq 0$, there is a polynomial $\mathcal{P}_{L,\lambda,b}$ with the following property. Fix an activation satisfying (1), an input dimension $d \geq 1$, and a Boolean target admitting a depth-L, width-$C \geq 1$ implementation with individual weights bounded by $B \geq 1$ and strict signed output margin $1/2$ at every input. For every input law D on the cube and every $0 < \varepsilon, \delta < 1$,*

$$n \geq \frac{2}{\varepsilon}\left[\mathcal{P}_{L,\lambda,b}(d, C, B, \log(1/\varepsilon)) + \log(1/\delta)\right]$$

implies, with training-sample probability at least $1 - \delta$,

$$R_D^{\mathrm{sign}}(\Pi_{S,N,t}) \leq \varepsilon \qquad \textit{simultaneously for every } N \geq C \textit{ and } 0 < t \leq 1.$$

All prior and posterior conditioning events have positive probability. The polynomial is uniform over the activations satisfying the displayed bounds. It includes every $\gamma \geq 1$ and the strong-prior limit at each fixed width, through $t = N^{(1-\gamma)/2}$ and $t \downarrow 0$, respectively.

As before, applying the theorem at accuracy $\varepsilon/3$ gives correct-label posterior probability strictly above $2/3$ outside test mass at most ε. The teacher must use the learner's activation; this does not assert a universal same-width circuit compiler for every Lipschitz function. The polynomial degree in this appendix may grow with L, in contrast with Theorem 1 under its additional support restriction.

A.1 A common capacity bound with offsets

Put $H = L - 1$ and retain the weighted energy

$$\kappa_N(w) = \frac{d}{N}\left\|W^1\right\|_F^2 + \sum_{s=2}^{H} \|W^s\|_F^2 + N\,\|v\|_2^2\,.$$

For counting only, allow each hidden neuron to use its own activation with Lipschitz constant at most λ and offset at most b. Let $\mathcal{F}_b(K)$ contain all Boolean functions having a strict margin-$1/2$ realization of energy at most K in this auxiliary class, at any finite equal hidden width. The actual learner still uses its single fixed activation.

Lemma 8 (Offset capacity)**.** *For $K \geq 0$, define*

$$\mathcal{B}_b(K) = \left\lceil \left[16H\lambda^H \sqrt{d+1} \left(\frac{(1+b^2)K + H}{L} \right)^{L/2} \right]^3 \right\rceil \log(2^d + 1).$$

Then $\log |\mathcal{F}_b(K)| \leq \mathcal{B}_b(K)$.

Proof. Write $a_j^s = \phi_{s,j}(0)$ and $\psi_{s,j} = \phi_{s,j} - a_j^s$. For a width-$N$ realization put $u^s = (h^s - a^s)/\sqrt{N}$. The state $(u^s, 1)$ is computed by a zero-at-zero network of the same depth, using upper-neuron activations $q \mapsto \psi_{s,j}(\sqrt{N}q)/\sqrt{N}$ and the identity activation for its additional constant wire. Its first matrix consists of $W^1/\sqrt{N}$ and the row selecting the constant input. Its internal matrices and readout are

$$\begin{pmatrix} W^s & W^s a^{s-1}/\sqrt{N} \\ 0 & 1 \end{pmatrix}, \qquad (\sqrt{N} v^\top,\ v^\top a^H).$$

These are exact identities, not approximations. Since $\|a^s\|_2 \leq b\sqrt{N}$, the sum of squared Frobenius norms of the transformed layer matrices is at most $(1+b^2)\kappa_N + H$, using $d \geq 1$. Their product is therefore at most $[((1+b^2)K + H)/L]^{L/2}$.

The zero-at-zero product-norm bound proved in [3] applies to arbitrary neuronwise activations of Lipschitz constant λ: on q inputs of norm at most R, its empirical Rademacher complexity is at most $8H\lambda^H PRq^{-1/3}$ when the product of layer norms is at most P. Here $R = \sqrt{d+1}$. Shattering with strict margin $1/2$ forces that complexity to be at least $1/2$, giving the displayed VC bound. The Sauer bound on the 2^d-point cube proves the result.

The normalization by $\sqrt{N}$ is important: unnormalized centered states would introduce columns $W^s a^{s-1}$ with an avoidable width factor. The construction is used only to count functions; it is not an identification of Gaussian priors. □

A.2 Target probability and small-scale compression

The target-cylinder argument of [3] requires only Lipschitz bounds on differences once the teacher's states are bounded. Here the teacher states satisfy

$$M_1 \leq b + \lambda B(d+1), \qquad M_s \leq b + \lambda CBM_{s-1}.$$

Thus a common bound M is polynomial in d, C, B at fixed L, λ, b. Use balanced cloning into C nonempty groups at every $N \geq C$; this preserves the exact teacher function for any activation and uses all neurons, so it does not assume zero padding is harmless. Multiply the teacher readout by three.

For completeness, one may use

$$M = [1 + b + \lambda B(d + C + 1)]^H, \quad g_H = 2 + \sqrt{2\log(2H)}, \quad F = 1 + \lambda(\sqrt{2}CB + \sqrt{C} + g_H),$$

$$V = 3\sqrt{2}CB, \qquad E = \lambda[(d+1) + H\sqrt{C}\,M]F^{H-1}, \qquad \rho = [8((V+1)E + M)]^{-1}.$$

Constrain first-layer, teacher-population, and readout coordinates exactly as in the target-cylinder proof of [3], to radius ρt in coordinates with variance t^2. The free internal Gaussian bulk annihilates teacher-group-constant states. With conditional probability at least $1/2$, all its operator norms are at most $g_H t$. On that event the normalized state error obeys

$$e_1 \le \lambda\rho(d+1), \qquad e_s \le F e_{s-1} + \lambda\rho\sqrt{C}\, M, \qquad e_H \le \rho E.$$

Hence output error is at most $(V+1)e_H + \rho M \le 1/8$. The draw has the teacher's signs with margin greater than $1/2$.

There are ND_0 constrained coordinates, where $D_0 = d+2+(H-1)C$, and their squared centers sum to at most NB_0, with

$$B_0 = B^2[d(d+1) + (2H+36)C^2].$$

The scalar interval estimate

$$\Pr\{|Z-a| \le \rho t\} \ge \frac{2\rho}{\sqrt{2\pi}} \exp(-a^2/t^2 - \rho^2), \qquad Z \sim N(0,t^2),$$

therefore gives

$$\mu_{N,t}(U_{N,f}) \ge e^{-A_f N/(2t^2)}, \qquad A_f = 2\left[B_0 + D_0 \log\frac{e\sqrt{2\pi}}{2\rho} + \log 2\right]. \tag{9}$$

The offset changes the state and radius budgets, but does not change the Gaussian coordinates, independence, or cancellation of the free bulk. All budgets have polynomial upper bounds at fixed depth.

The Gaussian spectral tail likewise makes no activation assumption. On its usual good event the weighted first-layer and readout energies are at most $K \ge 1$, and each internal matrix is the sum of a retained matrix of Frobenius norm at most $\sqrt{K}$ and a discarded matrix of operator norm at most $a \le 1$. Set

$$R = \lambda(\sqrt{K}+1) \ge 2, \qquad U = 2b + \lambda\sqrt{2K}.$$

The original hidden RMS norms satisfy $M_s \le UR^{s-1}$ because $M_1 \le b + \lambda\sqrt{2K}$ and $M_s \le b + RM_{s-1}$. Indeed, summing the additive offsets gives $M_s \le R^{s-1}(b+\lambda\sqrt{2K}) + b(R^{s-1}-1)/(R-1) \le UR^{s-1}$. For original versus truncated states,

$$e_1 = 0, \qquad e_s \le \lambda\sqrt{K} e_{s-1} + \lambda a M_{s-1} \le \lambda a (s-1) U R^{s-2}.$$

Taking

$$a_K = \min\{1, [16\lambda(H+1)\sqrt{K}(1+U)R^H]^{-1}\}$$

makes the readout discrepancy smaller than $1/8$. Doubling the readout of the truncated network preserves signs and strict margin $1/2$, at energy at most $(L+3)K$. The same Gaussian tails as in [3] then imply

$$\mu_{N,t}\left(\bigcup_{g \notin \mathcal{F}_b((L+3)K)} U_{N,g}\right) \le L e^{-NK/(8t^2)} \tag{10}$$

whenever $t \le a_K/8$ and $t^{-2} \ge 8(d+1)$. Here a_K^{-1} is polynomially bounded at fixed depth. When $H = 1$ there are no internal matrices to truncate and the discrepancy is zero.

A.3 The mean-field endpoint with offsets

This is the only part of the extension where an unmodified normalization would fail. We give the replacement and then check every use of the activation in the existing endpoint proof.

Let Q be a positive semidefinite matrix on $m = 2^d$ inputs, with diagonal at most $M \geq 1$, and write $\mathcal{H}_Q$ for its finite-dimensional reproducing-kernel space. The Gaussian projection lemma of [3] provides, for independent $Z_i \sim N(0, Q)$,

$$Z_i = a_i + u_i, \quad \|a_i\|_\infty \leq B_c := \sqrt{2M \log(4m)}, \quad \delta_i = \|u_i\|_Q ,$$

with

$$\Pr\left\{ N^{-1} \sum_i \delta_i^2 > A \right\} \leq e^{-Nq}, \qquad A = 4(q + \log 2).$$

For $Q = 0$ take $a_i = u_i = 0$. Keep its smooth cutoff r_T, which is one on $\|z\|_\infty \leq T$ and zero outside $\|z\|_\infty < \sqrt{5}T$. Put $w_i = 1 - r_T(Z_i)^2$.

Lemma 9 (Tail atoms with bounded offsets)**.** *Suppose $N^{-1} \sum_i \delta_i^2 \leq A$ and $T \geq \max\{1, 2B_c, b/\lambda\}$. The empirical activation covariance differs in maximum norm from*

$$\frac{1}{N} \sum_i r_T(Z_i)^2 \phi(Z_i)\phi(Z_i)^\top + \frac{1}{N} \sum_i w_i \phi(u_i)\phi(u_i)^\top$$

by at most $14\lambda^2 B_c MA/T$. The second term has an exact representation

$$\sum_i c_i \psi_i(v_i)\psi_i(v_i)^\top, \qquad \sum_i c_i \leq A + 1, \quad \|v_i\|_Q \leq 1, \quad \mathrm{Lip}(\psi_i) \leq \lambda, \quad |\psi_i(0)| \leq b.$$

The next covariance diagonal is at most $2(b + \lambda B_c)^2 + 2\lambda^2 MA$.

Proof. When $w_i > 0$, $\sqrt{M}\delta_i \geq T - B_c \geq T/2$. Consequently

$$N^{-1} \sum_i w_i \leq 4MA/T^2, \qquad N^{-1} \sum_i w_i \delta_i \leq 2\sqrt{M}A/T.$$

Use $|\phi(Z_i)_x - \phi(u_i)_x| \leq \lambda B_c$ and $|\phi(u_i)_x| \leq b + \lambda\sqrt{M}\delta_i$. The outer-product error is bounded by

$$\frac{4\lambda^2 B_c MA}{T} + \frac{8\lambda B_c bMA + 4\lambda^2 B_c^2 MA}{T^2} \leq \frac{14\lambda^2 B_c MA}{T}.$$

For the exact representation put

$$s_i = \sqrt{1 + \delta_i^2}, \qquad v_i = u_i/s_i, \qquad \psi_i(q) = \phi(s_i q)/s_i, \qquad c_i = w_i s_i^2/N.$$

Then $s_i \geq 1$ proves all assertions, including $\sum_i c_i \leq 1 + A$. This normalization has no division by zero. If $\delta_i = 0$, then $w_i = 0$; when $Q = 0$, the whole tail is empty and the bulk retains the actual constant feature $\phi(0)$. Finally $|\phi(Z_i)_x| \leq b + \lambda B_c + \lambda\sqrt{M}\delta_i$ gives the diagonal bound. □

The stable RKHS certificate used in [3] is unchanged: for $\|v\|_Q \leq 1$ and $\eta > 0$, it gives a vector β such that

$$\|Q\beta - v\|_\infty \leq \eta, \qquad \|\beta\|_1 \leq (4\eta)^{-1}.$$

For a mixture of the atoms above, put $A' = A + 1$ and $R_A = b + \lambda\sqrt{M}$. Every atom has absolute values at most R_A. For target error $0 < \xi \le 1$, choose

$$\eta = \frac{\xi}{64A'\lambda(R_A + \lambda + 1)}, \qquad n_A = \left\lceil 128(A')^2 R_A^4 \xi^{-2} \log(4m^2) \right\rceil.$$

Sample atoms with probabilities c_i/A'; fill the remaining probability with vector zero *and the identically zero activation*. Hoeffding's inequality gives an empirical covariance approximation within $\xi/4$. If $\|Q' - Q\|_{\max} \le \eta^2$, the certificates satisfy $\|Q'\beta - v\|_\infty \le 5\eta/4$. The activation error is at most $5\lambda\eta/4$, and each outer-product error at most $2R_A(5\lambda\eta/4) + (5\lambda\eta/4)^2$. The choice of η makes its total mixture contribution less than $\xi/8$. Thus the stable finite-atom realization continues to work with polynomial budgets. For a weighted feature realization $Q' = \sum_\ell a_\ell h_\ell h_\ell^\top$, its incoming weights remain

$$W_{j\ell} = a_\ell \sum_y \beta_j(y) h_\ell(y), \qquad |W_{j\ell}| \le a_\ell \|h_\ell\|_\infty / (4\eta).$$

The dummy activation must be zero: using $\phi(0)$ for the dummy atom would insert a spurious constant covariance.

The bounded bulk and its stability. Mollifying ϕ at scale $\tau \le 1$ changes it by at most $\lambda\tau$, retains first derivative bound λ, and gives second derivative bound $O(\lambda/\tau)$. None of these facts uses $\phi(0) = 0$. For the cutoff features $b_x(z) = r_T(z)\phi_\tau(z_x)$ and the chosen $T \ge b/\lambda$, the same entrywise derivative-sum bounds hold, with larger universal constants:

$$B_0 = O(\lambda T), \qquad B_1 = O(\lambda), \qquad B_2 = O(\lambda[\log(2m)/T + \tau^{-1}]).$$

Indeed the additional offset terms in the first and second derivative bounds are $O(b/T)$ and $O(b\log(2m)/T^2)$. The smoothing covariance error is $O(\lambda^2 T\tau)$.

The sparse exponential-tilt certificate and its Gaussian covariance stability estimate in [3] apply to arbitrary bounded C^2 features. They do not assume a zero activation offset. For clarity, with observables $F_{xy} = b_x b_y$, derivative budgets $F_0 = B_0^2$, $F_1 = 2B_0B_1$, $F_2 = 2B_1^2 + 2B_0B_2$, and tilt $\|\Gamma\|_{1,\mathrm{ent}} \le G$, their covariance map has modulus

$$C_G = \tfrac{1}{2}\{F_2 + 2GF_1^2 + 2F_0(GF_2 + G^2F_1^2)\}.$$

It is polynomial in the indicated budgets, including for singular covariances. The proof differentiates the normalized tilted expectation; it does not pay an exponential reciprocal partition-function bound.

Given a desired one-step error $\alpha \le 1$, take $q = 4K$, $A = 4(q + \log 2)$, and

$$T = \max\{1, 2B_c, b/\lambda, 256\lambda^2 B_c MA/\alpha\}.$$

Choose smoothing scale with reciprocal polynomial in T, α^{-1}, then $G = 16q/\alpha$ and the sparse certificate size $\max\{1, \lceil G^2F_0^2\rceil\}$. The original certificate proof has exceptional mass at most e^{-NK} above a polynomial width threshold. Combine its error $\alpha/16$, smoothing error $\alpha/16$, and the tail error from Lemma 9; use the atom construction with $\xi = \alpha/8$. Take the incoming covariance tolerance at most $\min\{1, \eta^2, \alpha/[16(1 + C_G)]\}$.

The bounded bulk is realized by finitely many tilted Gaussian rows, evaluating the *actual* activation ϕ and putting the cutoff scalars into their covariance weights. Its feature bound is $b + \lambda\sqrt{5}T = O(\lambda T)$. The Gaussian-coordinate intersection argument of [3] bounds all sampled

incoming coefficients polynomially, since the square root of the tilt log-density bound enters the coordinate cutoff. The expectation and sampling errors can each be made at most $\alpha/16$ with polynomially many rows. All errors total less than α.

Thus the one-step construction has the same two exceptional events, polynomial width and coefficient bounds, and a tolerance whose reciprocal is polynomial in $K, M, \log(2m), \alpha^{-1}$ at fixed λ, b. The physical hidden-value recursion becomes $B_h^{\mathrm{new}} \le b + \lambda r B_W B_h$, where r is the previous finite feature count and B_W the incoming weight bound. Its fixed-depth iteration remains polynomial. Padding uses the identically zero *auxiliary* activation, which belongs to the bounded-offset class; it does not introduce unused original-ϕ neurons with nonzero values.

Iteration and the final readout. Conditional on all previous layers, the next preactivation rows are independent centered Gaussians with covariance $t^2 Q_{s-1}$, where

$$Q_s(x,y) = N^{-1} \sum_i h_i^s(x) h_i^s(y), \qquad Q_0(x,y) = d^{-1} \bar{x} \cdot \bar{y}.$$

These are raw Gram matrices, not centered hidden covariances; hidden means therefore cause no change to this exact identity. Iterate the diagonal bound in Lemma 9, starting from $\max Q_0(x,x) \le 2$. Work backward from final covariance error $1/(128K)$ using the reciprocal-polynomial one-step tolerances, then reconstruct forward. Since H is fixed and $\log m = d \log 2$, all resulting widths, coefficient bounds and physical feature values are polynomial in d, K. The readout energy has law $t^2 \chi_N^2 / N$; except on mass e^{-NK} it is at most $8K$. The original output then has RKHS norm at most $\sqrt{8K}$. Its unchanged certificate and the final covariance approximation give output error at most $1/4$; doubling the reconstructed readout recovers its signs with strict margin $1/2$.

All reconstructed activations have offset at most b and Lipschitz constant at most λ: bulk neurons use ϕ, tail neurons use the normalized ψ_i, and dummy/padding neurons use zero. Bounding physical widths and weights bounds their weighted energy. Consequently there are polynomials $P_{L,\lambda,b}$ and $J_{L,\lambda,b}$ such that

$$N \ge P_{L,\lambda,b}(d,K) \quad \Longrightarrow \quad \mu_{N,t}\left(\bigcup_{g \notin \mathcal{F}_b(J_{L,\lambda,b}(d,K))} U_{N,g} \right) \le (2H+1) e^{-NK}. \tag{11}$$

The conditional union bound accounts for two events per hidden layer and one readout event. This holds uniformly for $0 < t \le 1$: covariance multiplication by t^2 does not enlarge diagonal or approximation budgets, bulk incoming weights acquire a factor t, atom incoming weights a factor t^2, and readout tails only decrease. The offset itself is still bounded by b.

A.4 One finite class for every width and scale

Proof of Theorem 7. Use A_f from (9) and set

$$\begin{aligned}
K_s &= 4A_f + 8\log(2L/\varepsilon), & t_* &= \min\{a_{K_s}/8, [8(d+1)]^{-1/2}\},\\
K_m &= 2 + A_f/t_*^2 + \log(2(2H+1)/\varepsilon), & N_* &= \lceil \max\{1, P_{L,\lambda,b}(d,K_m)\} \rceil,\\
D_* &= d + 2 + (L-2)N_*, & J_0 &= 2A_f + 2D_* \log 2 + 4\log(2/\varepsilon),\\
J_* &= (L+3)K_s + J_{L,\lambda,b}(d,K_m) + J_0.
\end{aligned}$$

Increase the polynomial envelopes to be nonnegative if needed. Every quantity has a polynomial upper bound in $d, C, B, \log(1/\varepsilon)$ at fixed L, λ, b, including t_*^{-1}.

Let $\mathcal{F}_* = \mathcal{F}_b(J_*)$. For $t \le t_*$, divide the spectral bound (10) by the target probability to get

$$\frac{\mu_{N,t}(\bigcup_{g\notin\mathcal{F}_*} U_{N,g})}{\mu_{N,t}(U_{N,f})} \le L\exp[-N(K_s/8 - A_f/2)/t^2] \le \varepsilon/2.$$

For $t \ge t_*$ and $N \ge N_*$, the endpoint bound (11) gives the same conclusion, since its ratio is at most $(2H+1)\exp[-N(K_m - A_f/(2t^2))]$. For $C \le N < N_*$, the parameter count is $p = N(d+2) + (L-2)N^2 \le ND_*$. Independently of the activation, $\kappa_N = (t^2/N)\chi^2_p$ under the raw prior. Its exponential moment at $N/(4t^2)$ is $2^{p/2}$, so

$$\frac{\mu_{N,t}\{\kappa_N > J_0\}}{\mu_{N,t}(U_{N,f})} \le \exp\left[-\frac{N}{t^2}(J_0/4 - A_f/2) + \frac{p}{2}\log 2\right] \le \varepsilon/2.$$

Every margin-separated draw of energy at most J_0 is itself in the auxiliary bounded-offset class. Thus the same relative-mass bound holds in every case.

Conditioning on the global margin cancels from probability ratios. Noiseless training always leaves the target in the posterior denominator, so posterior mass outside $\mathcal{F}_*$ is at most $\varepsilon/2$ on *every* training history. By Lemma 8, $\log|\mathcal{F}_*| \le \mathcal{B}_b(J_*)$. A finite-class union bound gives a sample event of probability at least $1-\delta$ on which every consistent member has true error at most $\varepsilon/2$, provided

$$n \ge \frac{2}{\varepsilon}[\mathcal{B}_b(J_*) + \log(1/\delta)].$$

This event depends only on one finite function class and therefore works for the entire family of widths and scales. Adding the discarded posterior mass proves the risk guarantee. A polynomial upper envelope of $\mathcal{B}_b(J_*)$ supplies $\mathcal{P}_{L,\lambda,b}$. The target-cylinder lower bound proves all normalizers positive. □

References

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